Plasma Arc Extraction Furnace
Class 23

Note: the figures below are conceptual reference illustrations of the design carried over from the source diagram. They are not formal patent-office-compliant line drawings — for an actual filing, commission clean, unlabeled orthographic line drawings (perspective, front, side, top, bottom views) from a patent draftsperson.

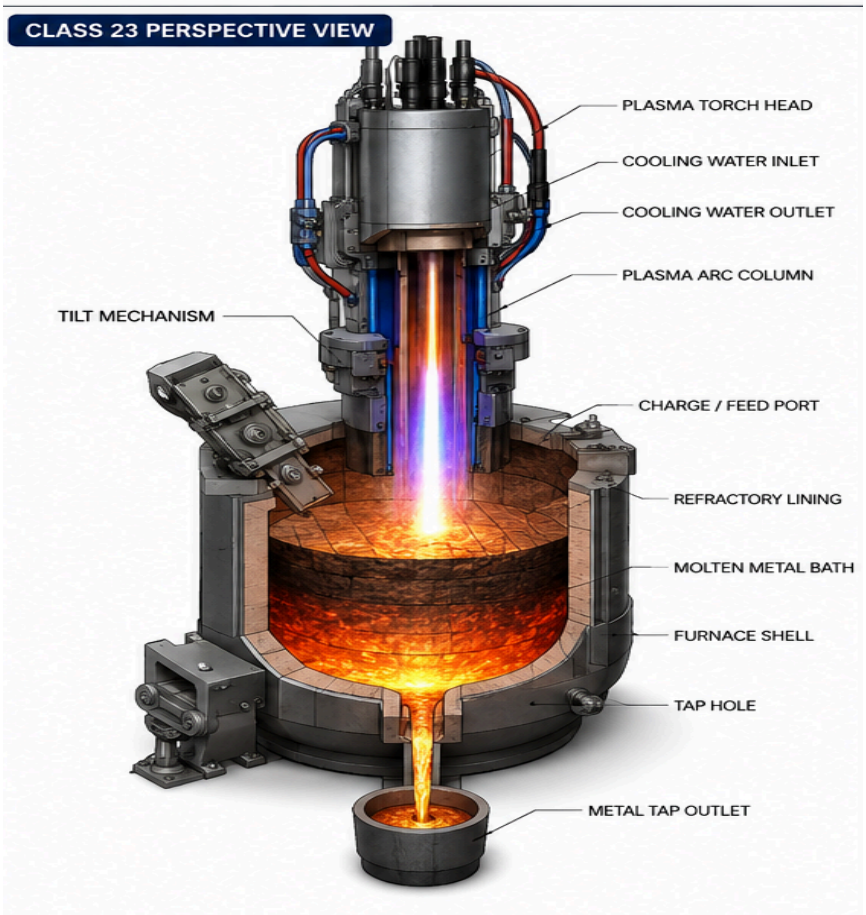


Figure 1: The perspective view

Figure 1 shows the overall perspective view of the Plasma Arc Extraction Furnace. The design consists of a high-temperature plasma torch placed at the centre of the furnace chamber. The torch produces a powerful ionized gas jet that travels downward into the furnace and helps melt the feed material and separate valuable metal from slag.

The furnace has several important parts, including a feed hopper, water-cooled torch nozzle, refractory-lined chamber, and two separate tap holes for removing slag and molten metal. These components are combined into one compact furnace structure, giving it a distinctive industrial appearance suitable for metal recovery and e-waste processing.

The main visual feature of the proposed design is the combined arrangement of the torch housing, feed hopper, and dual tap-hole furnace body. Together, these elements create a unique and recognizable shape that is visually different from conventional cupola and induction furnaces.

Plasma Arc Extraction Furnace
Class 23

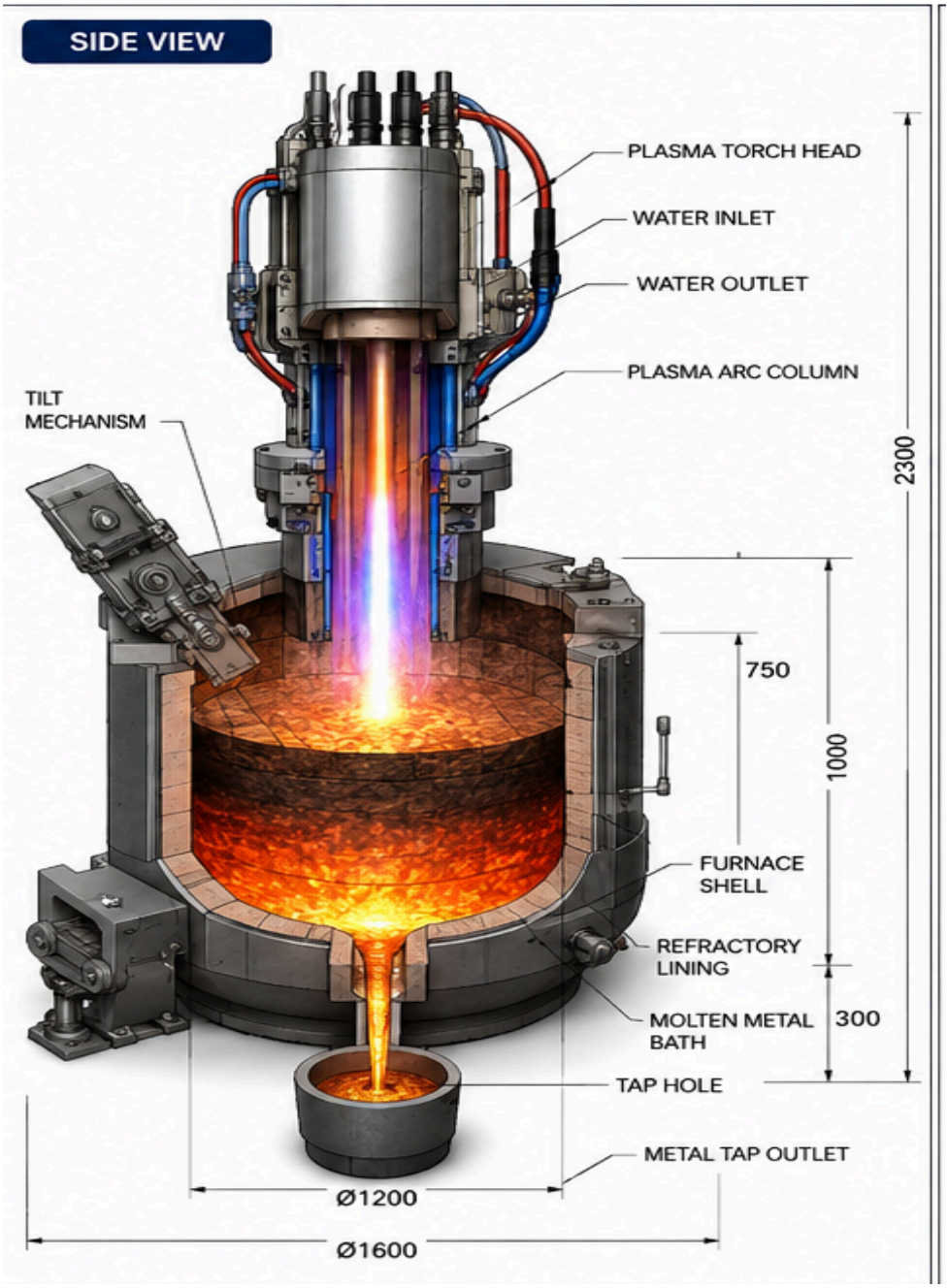


Figure 2: The side view

Figure 2 shows the side view of the Plasma Arc Extraction Furnace and explains how the main components are arranged around the furnace body. The side view includes the plasma torch, power electrode housing, gas inlet, and control panel.

The lower part of the furnace has a tapered shape, while the control panel is placed slightly to the side. The slag tap is positioned above the lower metal tap hole, making the two outlets clearly separate and easy to distinguish in the side view.

Plasma Arc Extraction Furnace
Class 23

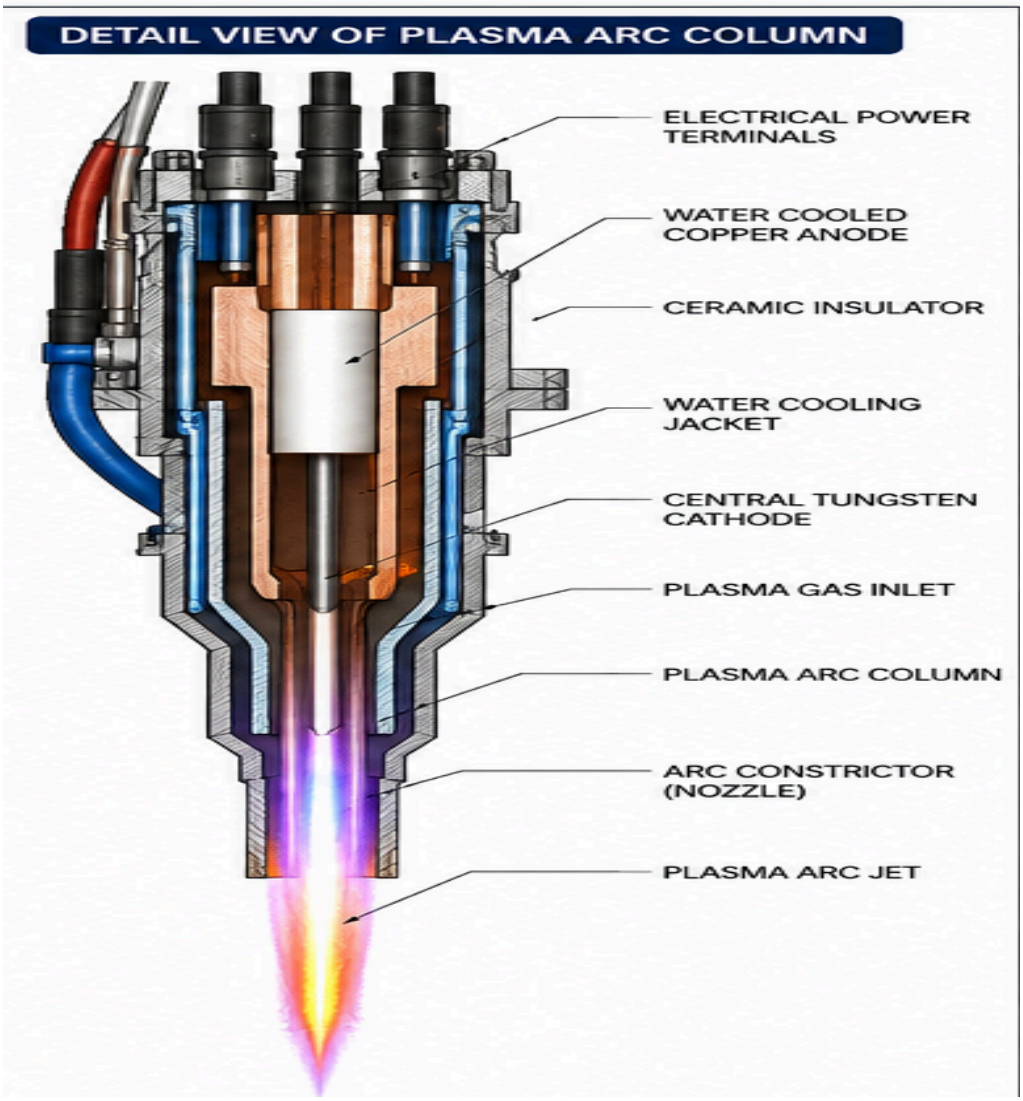


Figure 3: Detail view of the plasma arc column

Figure 3 shows a detailed view of the plasma arc column inside the furnace during operation. The plasma forms a long, bright jet that extends downward from the torch into the furnace chamber. This glowing plasma column is one of the main visual features of the furnace and gives the design a distinctive operating appearance.

Originality Statement

To the best of our knowledge, the design and concept presented in this application were developed independently. The proposed design combines the torch housing, feed hopper, dual tap-hole furnace body, and control unit into one integrated structure. This combination gives the plasma arc metal-extraction furnace a distinctive and novel ornamental appearance.